

Tarefa 1

Bioinformática Geral

Do sequenciamento ao alinhamento

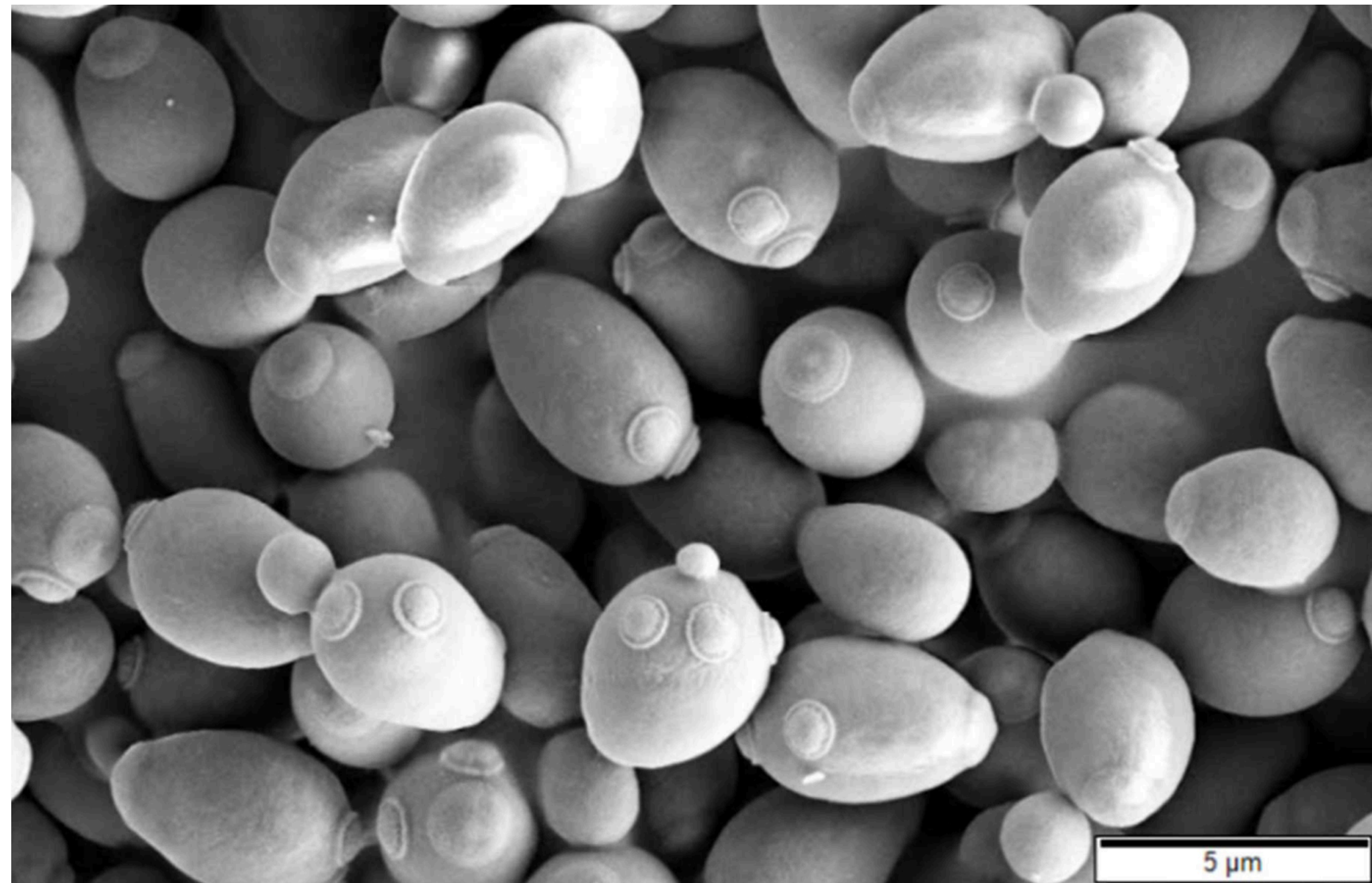
Gabriela Rodriguez, Henrique Macedo e Paulo Schreiner

Saccharomyces cerevisiae

12,1 Mb totais

16 cromossomos lineares

Ploidia variável



Escolher genoma de referência

Selected taxa						
Saccharomyces cerevisiae × Enter one or more taxonomic names						
Filters						
Download ▼ Select columns 1,756 Genomes Rows per page 20 ▼ 1-20 of 1,756 < >						
☐ Assembly	GenBank	RefSeq	Scientific name	Modifier	Action	
☐ R64 	GCA_000146045.2	GCF_000146045.2	Saccharomyces cerevisiae S288C	S288C (strain)	⋮	
☐ ASM2117220v1	GCA_021172205.1		Saccharomyces cerevisiae (brewer's yeast)	S288C-SK1-cross (strain)	⋮	
☐ ASM2350882v1	GCA_023508825.1		Saccharomyces cerevisiae (brewer's yeast)	CICC 1445 (strain)	⋮	
☐ ASM401491v1	GCA_004014915.1		Saccharomyces cerevisiae (brewer's yeast)	Makgeolli (strain)	⋮	
☐ ASM308665v1	GCA_003086655.1		Saccharomyces cerevisiae (brewer's yeast)	BY4742 (strain)	⋮	
☐ SX2 Assembly v2	GCA_903819175.2		Saccharomyces cerevisiae (brewer's yeast)	SX2 (strain)	⋮	
☐ ASM5731852v1	GCA_057318525.1		Saccharomyces cerevisiae (brewer's yeast)	WUT3 (strain)	⋮	
☐ ASM158042v1	GCA_001580425.1		Saccharomyces cerevisiae (brewer's yeast)	BCAnc (strain)	⋮	
☐ ASM5464329v1	GCA_054643295.1		Saccharomyces cerevisiae (brewer's yeast)	x303 (strain)	⋮	
☐ ASM5569733v1	GCA_055697335.1		Saccharomyces cerevisiae (brewer's yeast)	A154 (strain)	⋮	
☐ ASM2497295v1	GCA_024972955.1		Saccharomyces cerevisiae (brewer's yeast)	L261 (strain)	⋮	
☐ ASM2497293v1	GCA_024972935.1		Saccharomyces cerevisiae (brewer's yeast)	L261col5 max (strain)	⋮	

Escolher genoma de referência

Genome assembly R64 reference

[Download](#)

 [datasets](#)

[API](#)

[FTP](#)

Actions

NCBI RefSeq assembly

GCF_000146045.2 ⁽⁺⁾

Submitted GenBank assembly

GCA_000146045.2

Taxon

[Saccharomyces cerevisiae S288C](#)

Strain

S288C

Synonym

sacCer3

Assembly type

haploid

Submitter

Saccharomyces Genome Database

Date

Dec 17, 2014

Additional genomes

[Browse all Saccharomyces cerevisiae genomes \(1756\)](#)

BioProject

[PRJNA43747](#)

Third Party Annotation (TPA) genome maintained by SGD

Publications

Genetics 2022

[New data and collaborations at the Saccharomyces Genome Database: updated reference genome, alleles, and the Alliance of Genome Resources](#)

Engel, Stacia R, et al.

FEBS Lett 1998

[The complete sequence of the mitochondrial genome of Saccharomyces cerevisiae](#)

Foury, F, et al.



[View annotated genes](#)



[See in Genome Data Viewer](#)

Escolher SRA - reads para alinhar

<https://www.ncbi.nlm.nih.gov/sra/SRX21572403>[accn]

SRX21572403: Wyeast 3068, a Weihenstephan Weizen tetraploid *S. cerevisiae* ale strain frequently used to brew Hefeweizen style beers

1 ILLUMINA (NextSeq 500) run: 1.6M spots, 802.9M bases, 447.4Mb downloads

Design: sample taken from commercial wort

Submitted by: University of Washington

Study: Systematic profiling of Hefeweizen ale yeast across fermentation and repitching reveals global trends in protein dynamics during brewing

[PRJNA1011390](#) • [SRP457872](#) • [All experiments](#) • [All runs](#)

[show Abstract](#)

Sample:

[SAMN37218449](#) • [SRS18762164](#) • [All experiments](#) • [All runs](#)

Organism: [Saccharomyces cerevisiae](#)

Library:

Name: LO-06

Instrument: NextSeq 500

Strategy: WGS

Source: GENOMIC

Selection: RANDOM

Layout: PAIRED

Runs: 1 run, 1.6M spots, 802.9M bases, [447.4Mb](#)

Run	# of Spots	# of Bases	Size	Published
SRR25850990	1,605,740	802.9M	447.4Mb	2023-08-31

Preparar ambiente no mamba

```
mamba create -n bioinfo-geral  
mamba activate bioinfo-geral
```

```
conda config --add channels bioconda  
conda config --add channels conda-forge  
conda config --set channel_priority strict  
# Essa config é global e se aplica a todos ambientes editar o .condarc
```

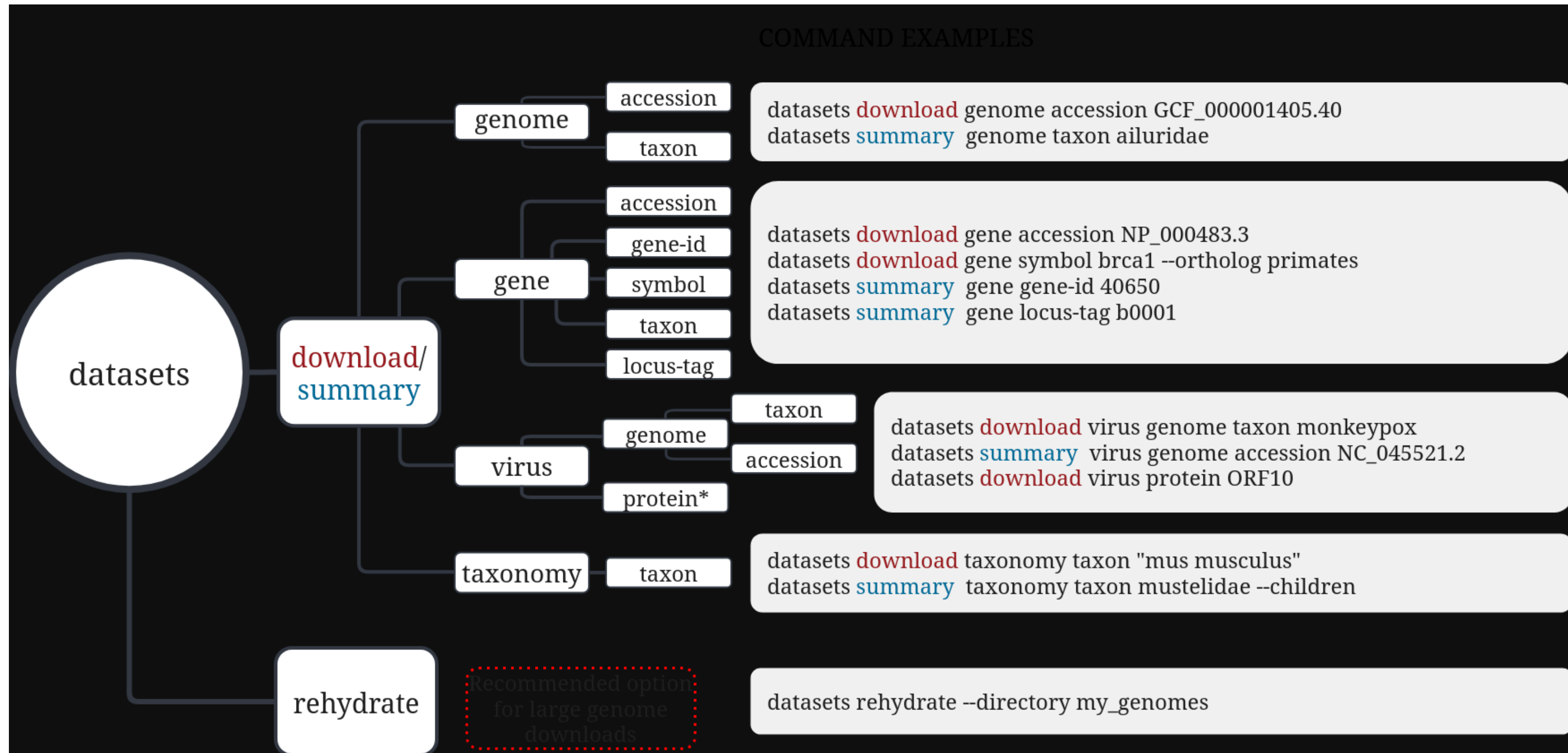
```
mamba install ncbi-datasets-cli sra-tools bwa samtools igv -y
```



Contém o fastq-dump

NCBI datasets CLI

<https://www.ncbi.nlm.nih.gov/datasets/docs/v2/command-line-tools/download-and-install/>



* virus protein: restricted to download of SARS-Cov-2 proteins

Baixar genoma de referência

```
datasets download genome accession GCF_000146045.2 --include genome --  
filename ref_genome.zip
```

```
unzip ref_genome.zip
```

```
tree
```

```
.  
├── README.md  
├── reads  
└── ref  
    └── ref_genome.fna
```


Baixar reads do SRA

```
fastq-dump -X 50000 -F --outdir reads --split-files SRR25850990
```

```
# --split-files faz com que as reads F e R venham em arquivos separados
```

```
tree
```

```
.
├── README.md
├── reads
│   ├── SRR25850990_1.fastq
│   └── SRR25850990_2.fastq
└── ref
    └── ref_genome.fna
```

Indexar genoma de referência

```
bwa index ref_genome.fna
```

```
# foram gerados os arquivos:
```

```
ls ref/
```

```
ref_genome.fna      ref_genome.fna.ann  ref_genome.fna.pac
```

```
ref_genome.fna.amb  ref_genome.fna.bwt  ref_genome.fna.sa
```

Alinhar as reads contra o genoma

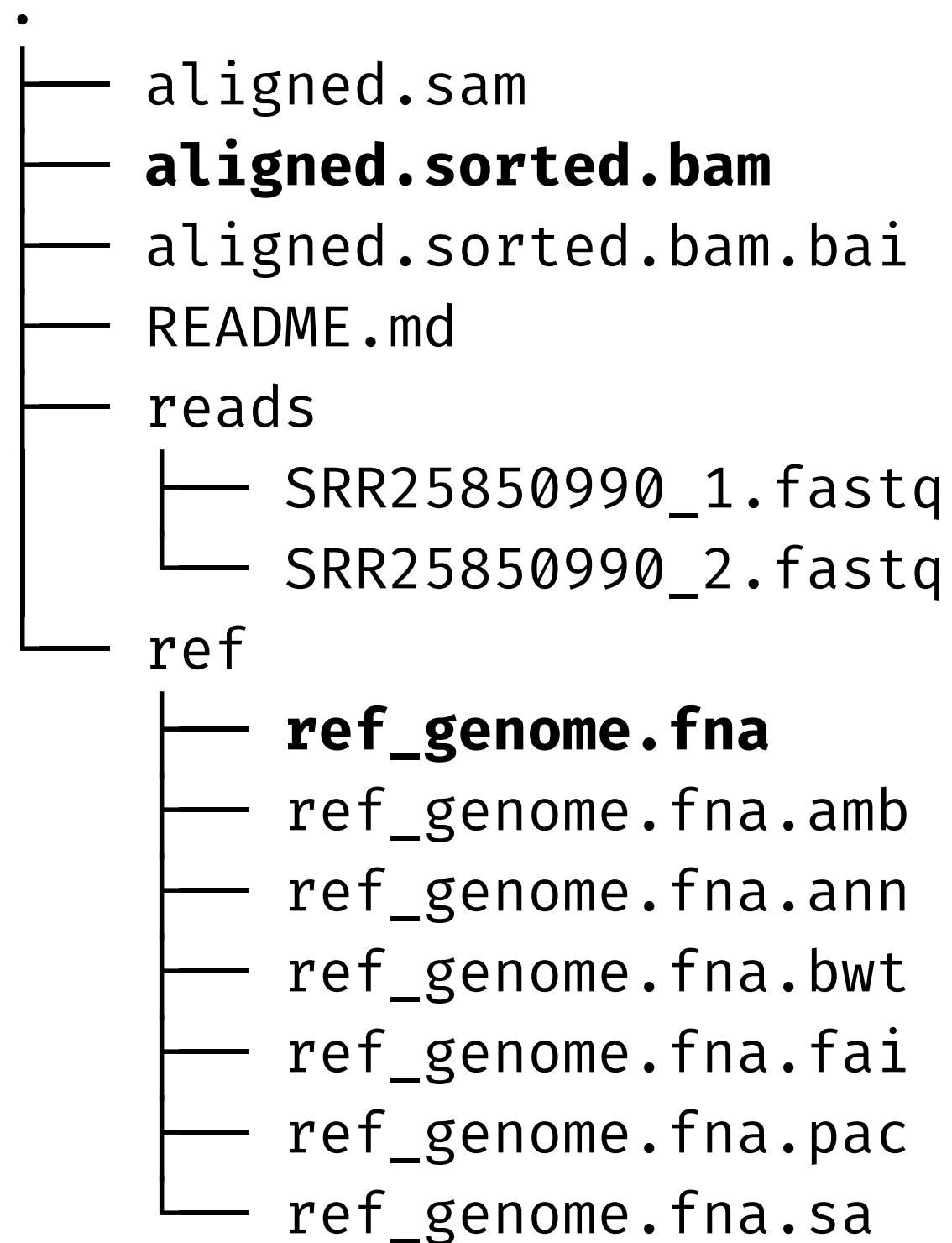
```
bwa mem -t 4 \  
ref/ref_genome.fna \  
reads/SRR25850990_1.fastq \  
reads/SRR25850990_2.fastq > aligned.sam
```

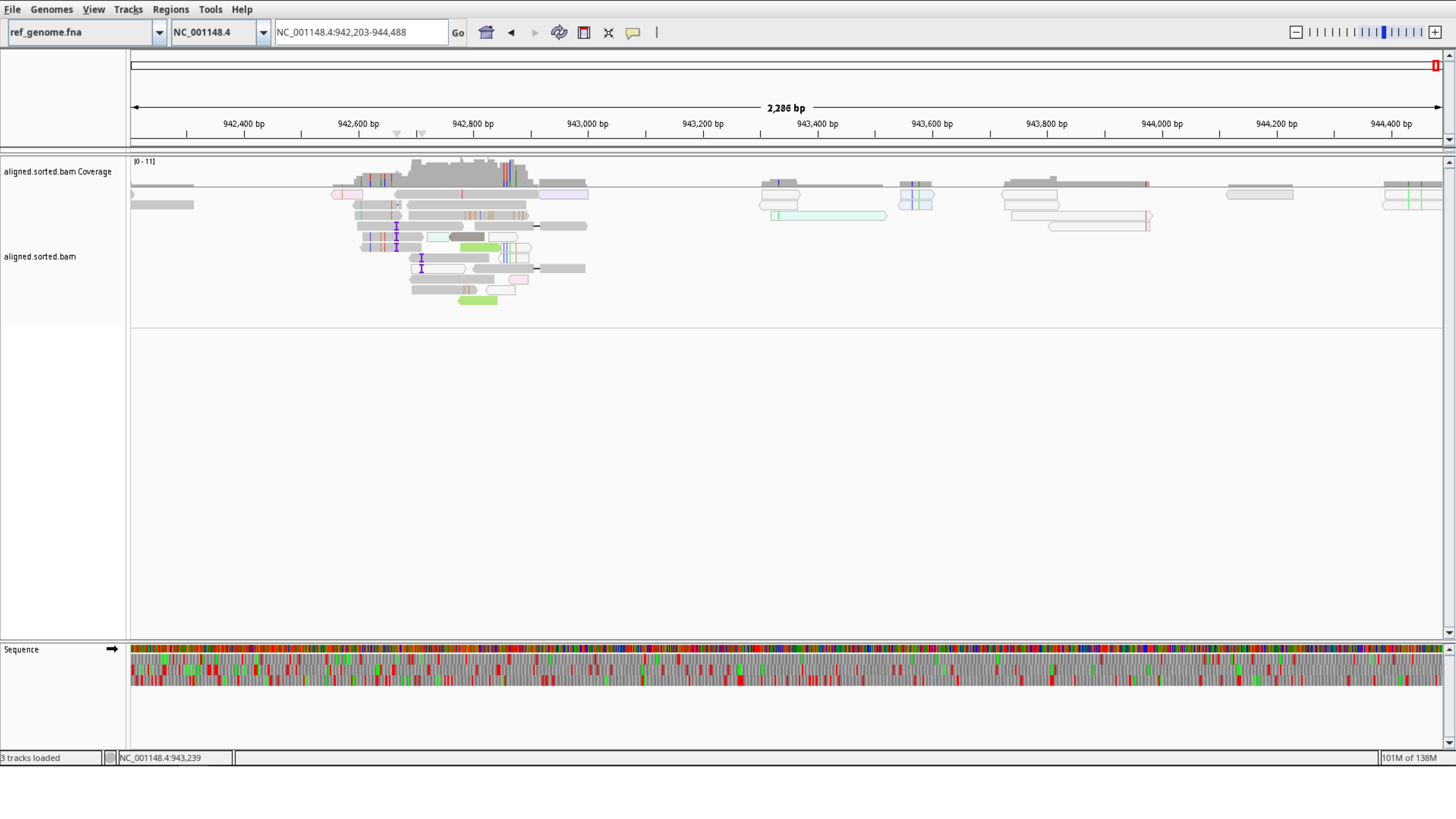
Gerar e indexar o BAM

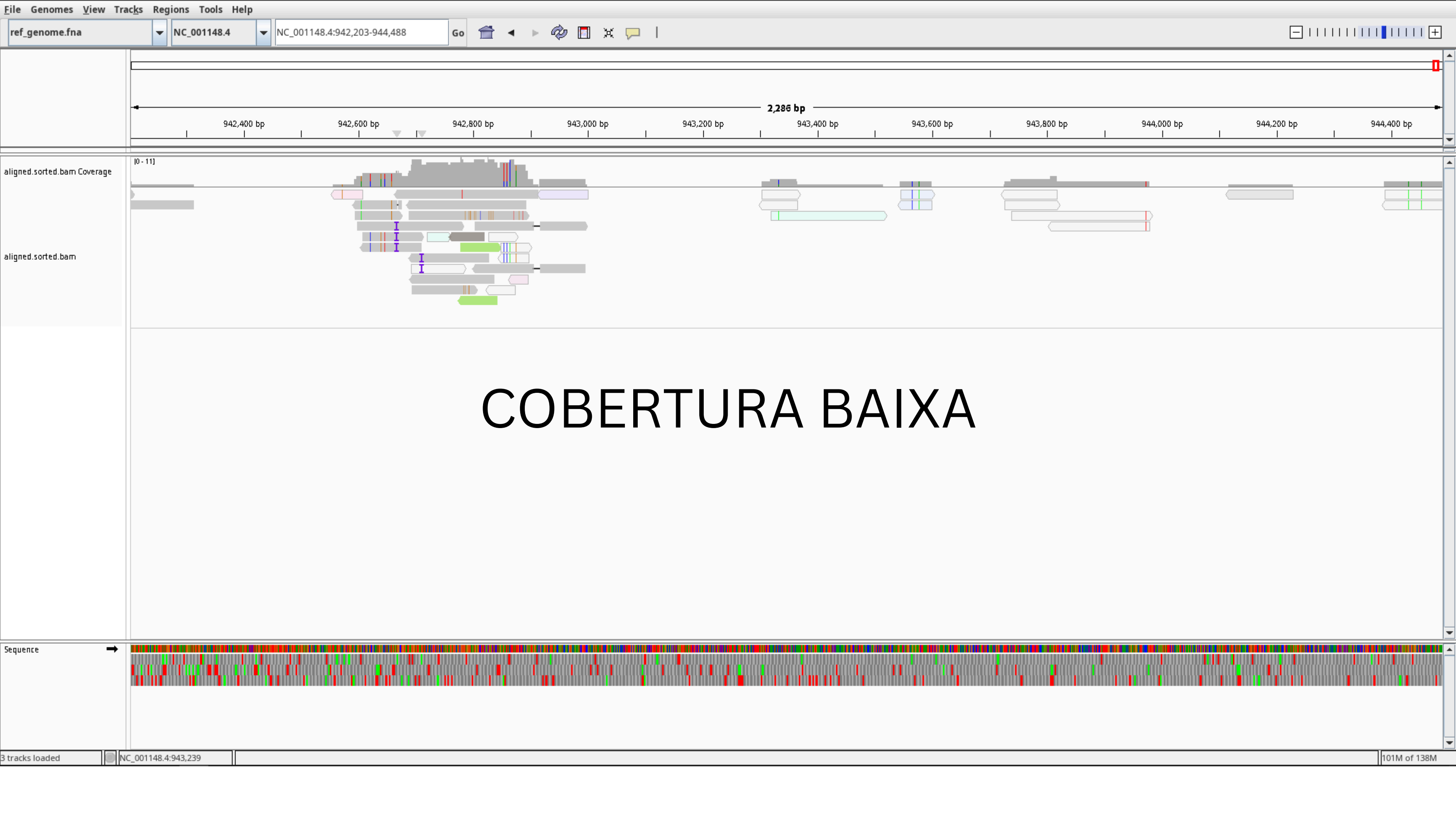
```
cat aligned.sam | samtools sort > aligned.sorted.bam
```

```
samtools index aligned.sorted.bam
```

Arquivos gerados







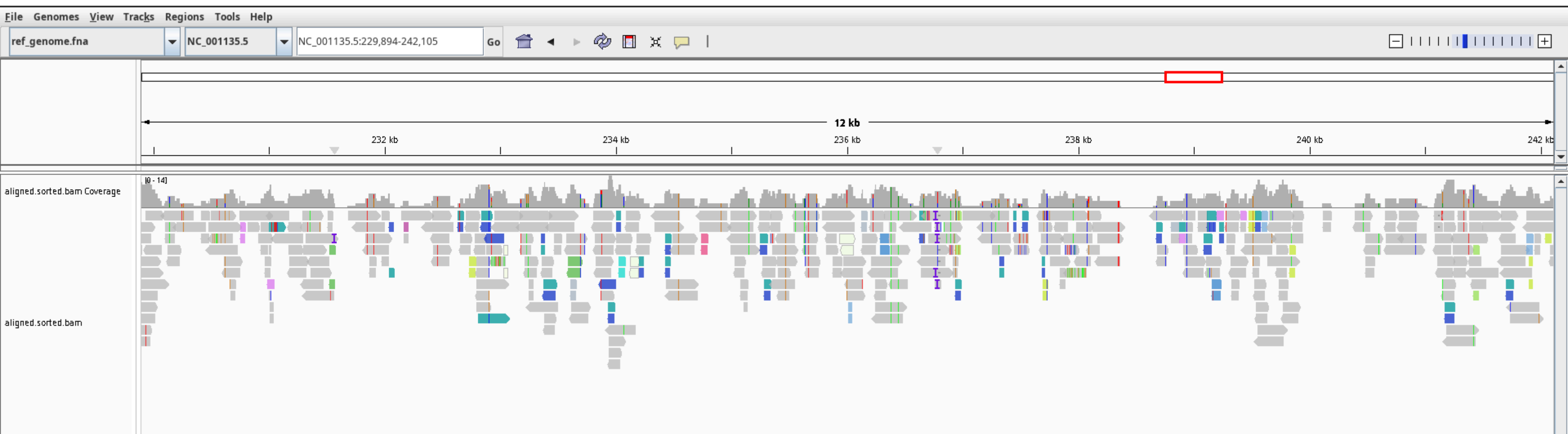
Baixar reads do SRA (de novo)

```
fastq-dump -X 200000 -F --outdir reads --split-files SRR25850990
```

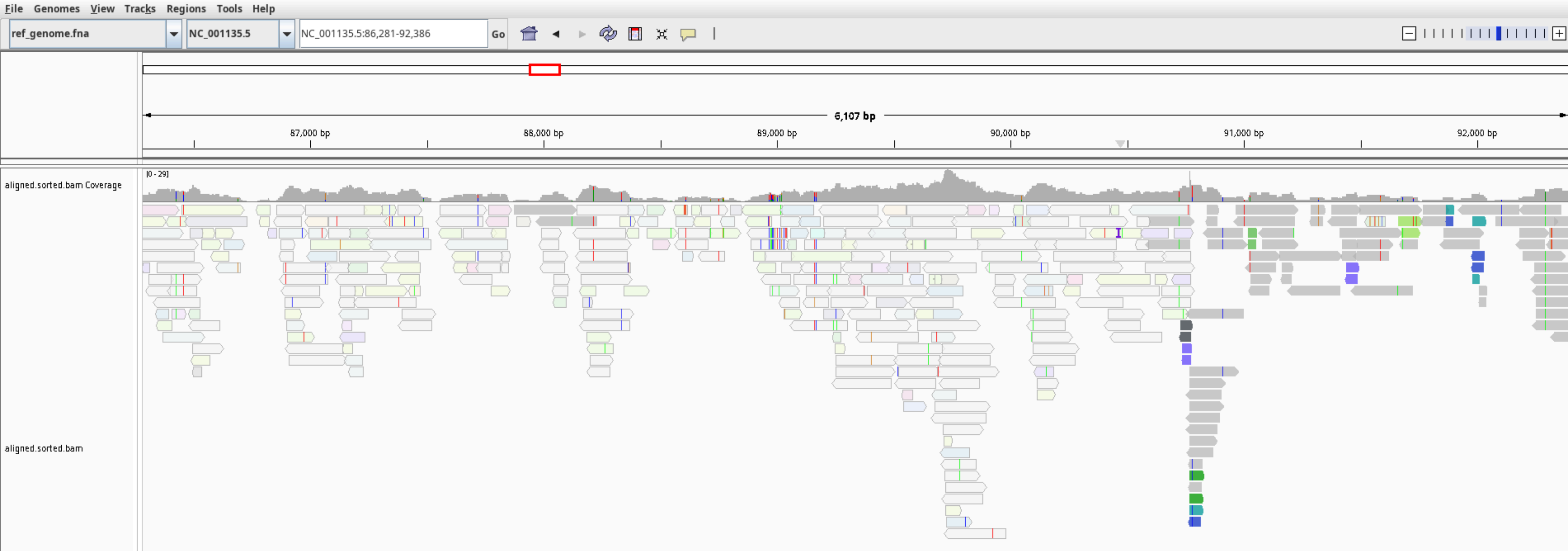
```
# --split-files faz com que as reads F e R venham em arquivos separados
```

```
tree
```

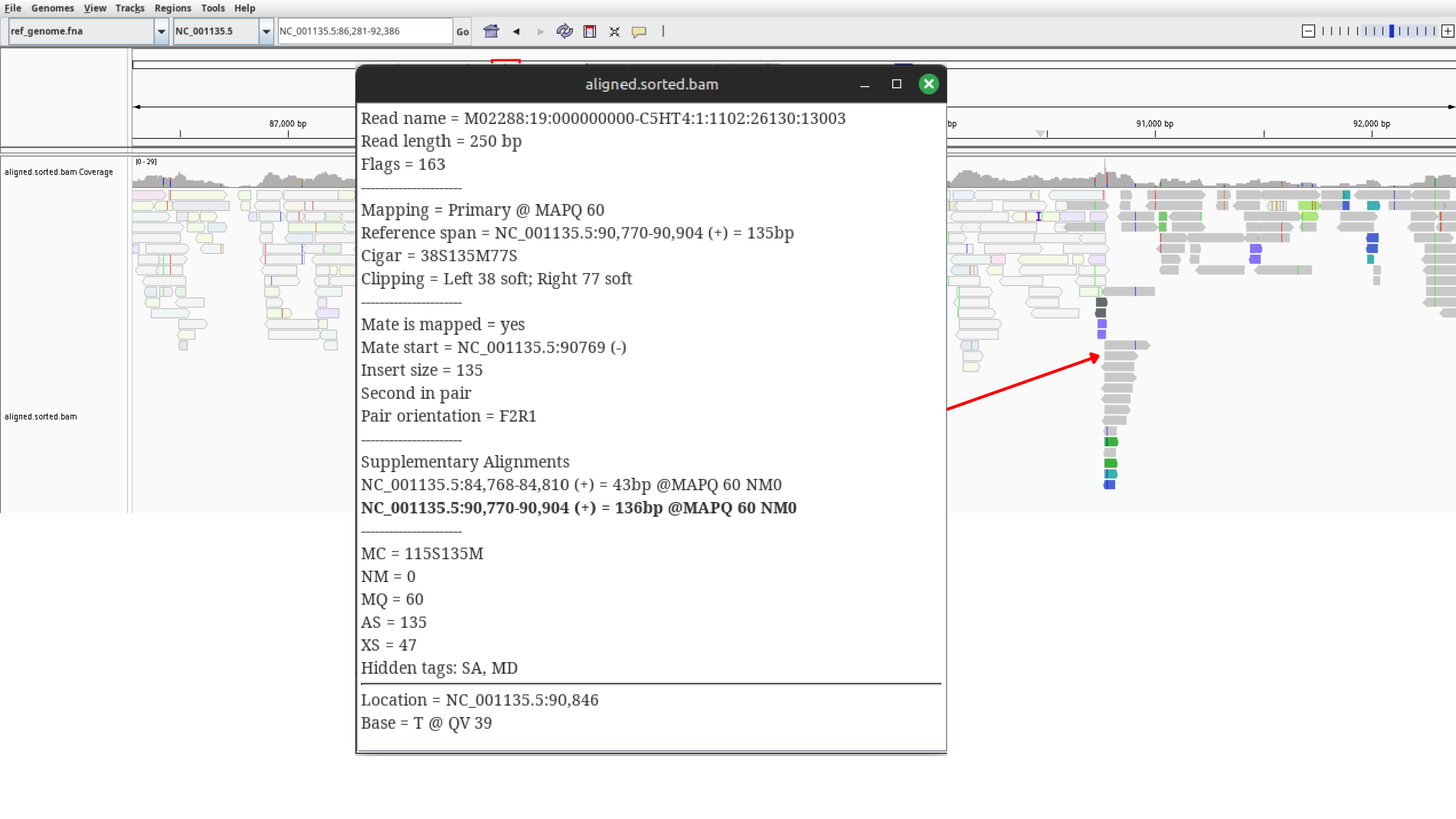
```
.
├── README.md
├── reads
│   ├── SRR25850990_1.fastq
│   └── SRR25850990_2.fastq
└── ref
    └── ref_genome.fna
```

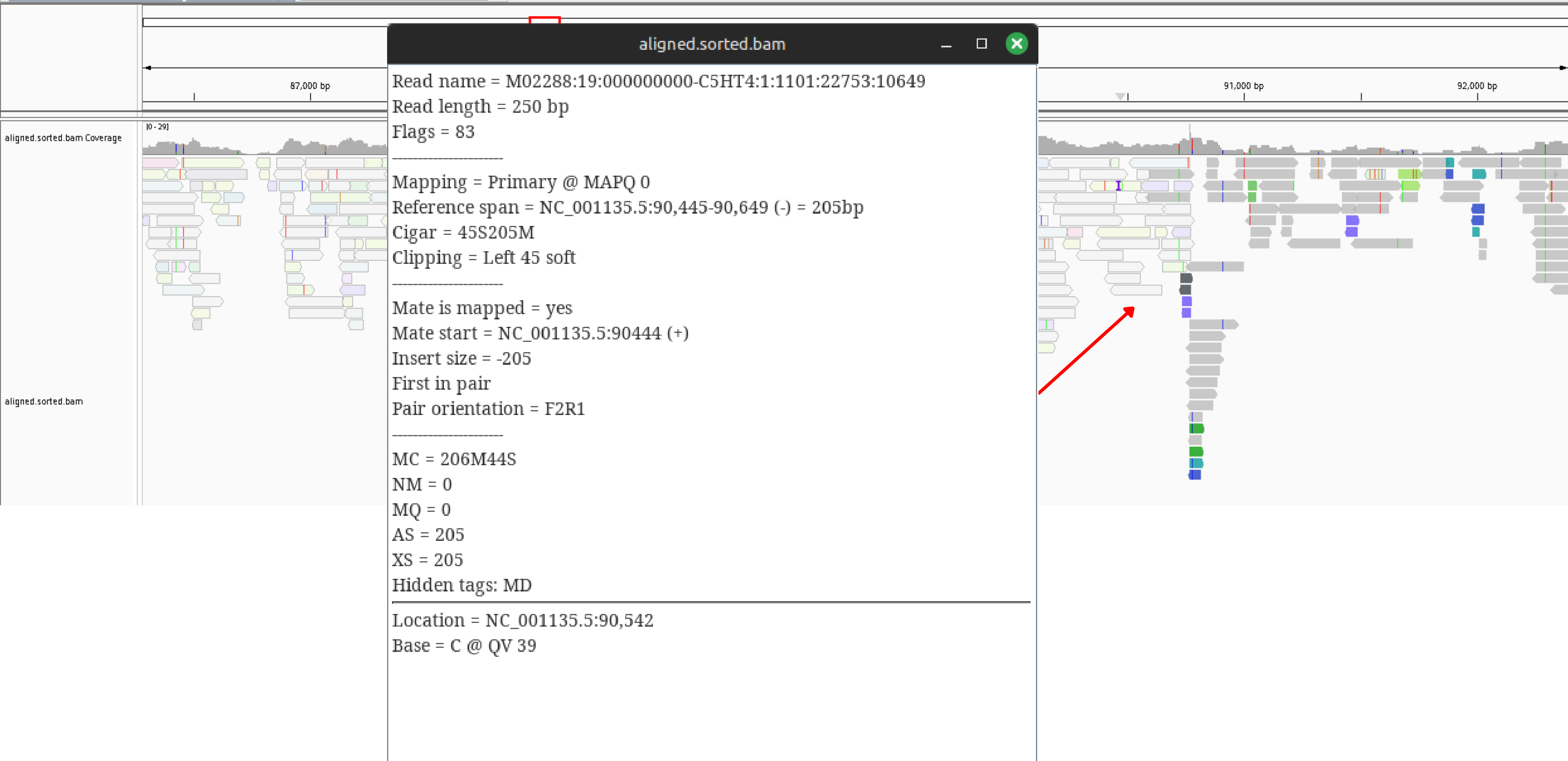


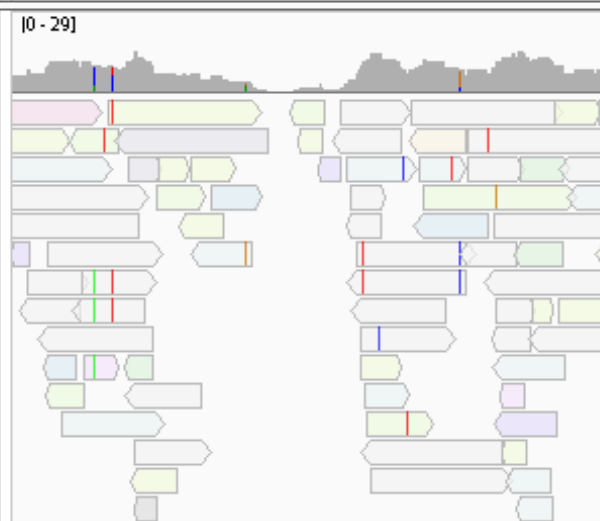
Melhor assim!



O que é essa região de reads de cor clara?



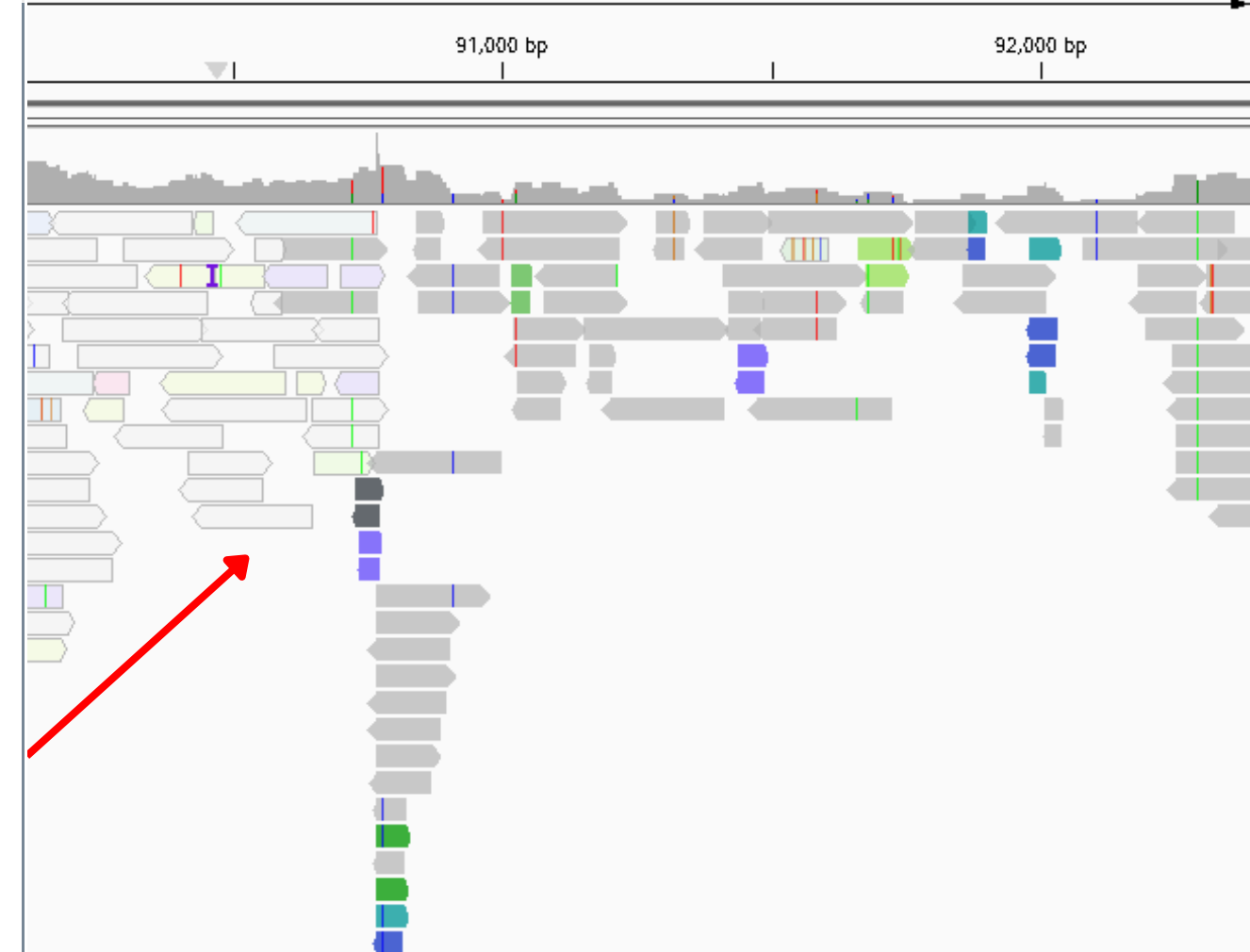




aligned.sorted.bam

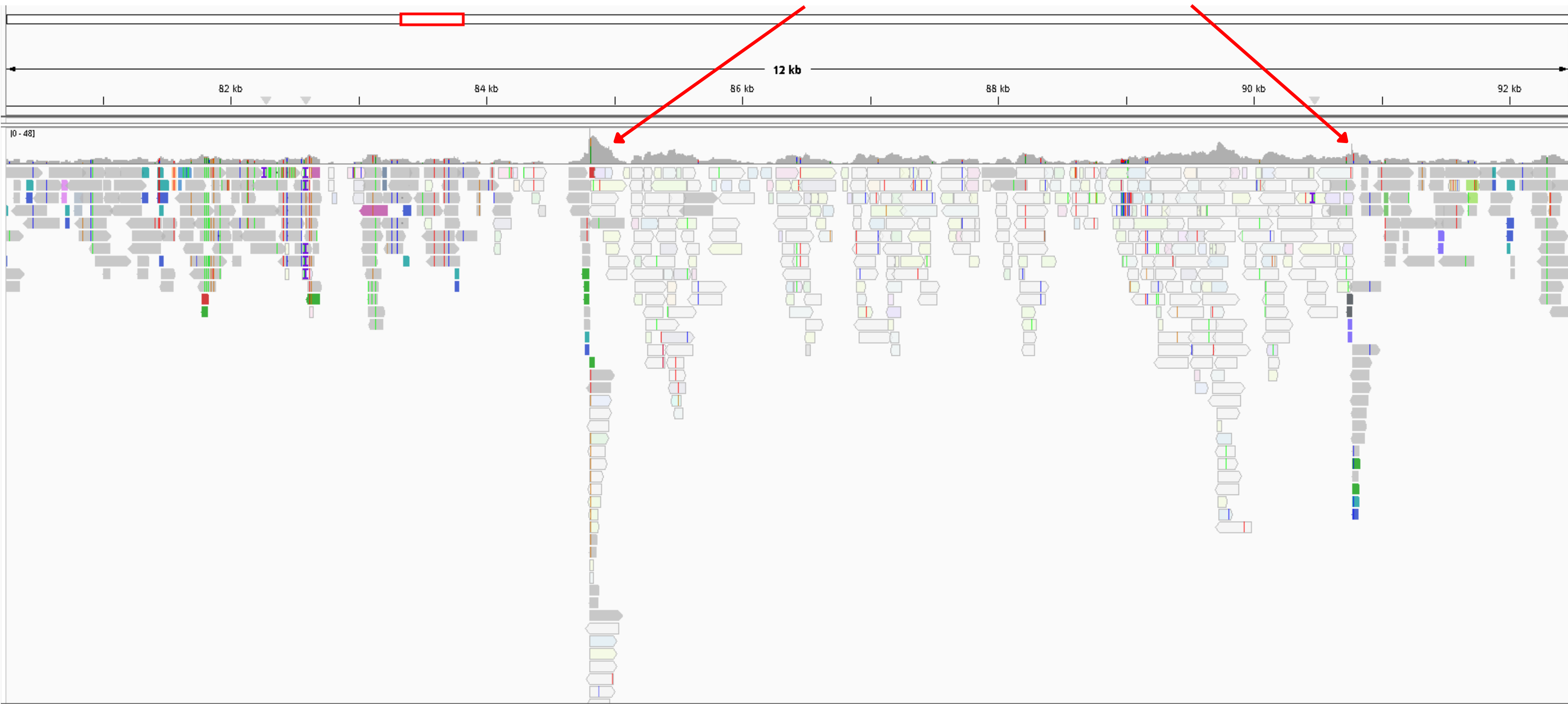
Base = C @ QV 39

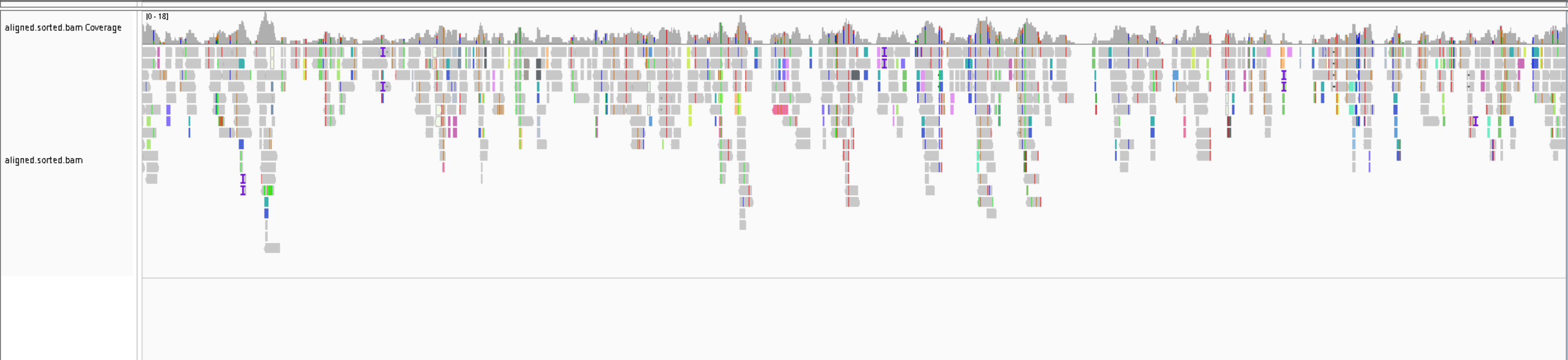
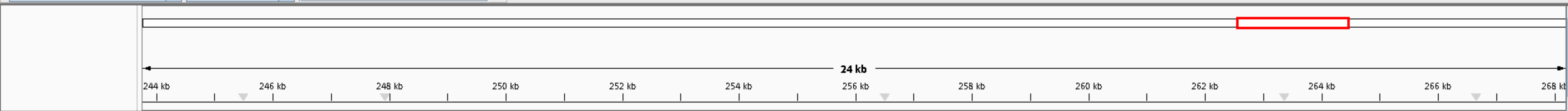
*Read length = 250pb
Cigar = 45S205M
↳ 45S + 205M diz que 205pb
alinham contra a
referência e 45 sofreram
soft-clipping.*



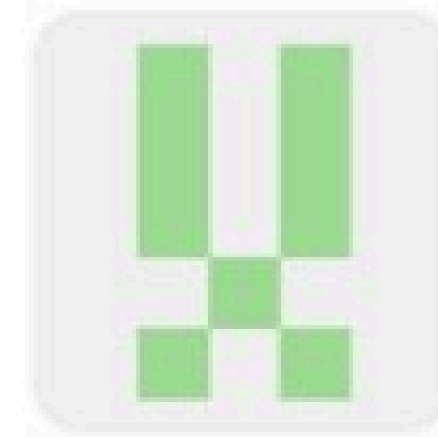
Variante estrutural

Pontos de quebra





PauloSchreiner/ **bioinformatica-geral-...**



1

Contributor



0

Issues



0

Stars



0

Forks



PauloSchreiner/bioinformatica-geral-ufrgs

Contribute to PauloSchreiner/bioinformatica-geral-ufrgs development by creating an account on GitHub.

